

Open-Source Visayas Grid Model User Manual

Introduction

The future of the Visayas Grid presents a significant area for study, with an estimated 24,750 MW increase in renewable energy generation by 2033 and a projected demand of 10,678 MW by 2050. Despite this rapid development, there is a limited availability of publicly accessible data and educational power system models specifically catered to the Philippines. This project addresses these challenges through the development of an open-source Visayas grid model using publicly available data from institutions such as the Independent Electricity Market Operator of the Philippines (IEMOP), Department of Energy (DOE), National Grid Corporation of the Philippines (NGCP), and the Wholesale Electricity Spot Market (WESM), as well as other online databases. In cases where data are unavailable, values are estimated or assumed based on standard engineering practices.

The model is implemented using **Pandapower**, a Python-based power system analysis tool, and is integrated with a graphical user interface (GUI) developed using Dear PyGui. It is designed to support **power flow analysis and optimal power flow simulations**. To reflect realistic operating conditions, the application includes three base cases corresponding to specific time periods:

- December 22, 2025 – 10:00 AM
- January 28, 2026 – 11:00 PM
- February 8, 2026 – 3:00 PM

Through the interface, users are able to visualize the network, perform simulations, and modify system parameters to observe their effects on grid behavior. It is important to note that this model is not intended to be a completely accurate representation of the Visayas Grid. Rather, it is designed to serve as an initial, open-source platform that can be used, improved, and expanded upon.

Overall, the model aims to provide a starting point for power system studies, supporting academic learning and research while encouraging further development of accessible power system tools for education in the Philippine context.

Features

The Open-Source Visayas Grid Model provides a set of core functionalities designed for learning and analysis:

- **Power Flow Analysis**

Performs steady-state AC power flow using the Newton-Raphson method. Results include bus voltages, voltage angles, line loading, and transformer loading.

- **AC Optimal Power Flow**

Following the power flow, the application runs an AC optimal power flow to determine least-cost generator dispatch subject to network and capacity constraints. Results include optimal MW per unit and Locational Marginal Prices (LMP in Php/MWh) at every bus. The LMP represents the marginal cost of supplying one additional MW of load at that location.

- **Network Modification**

Users can modify the network by (1) uploading a replacement CSV for any component in a chosen subgrid; (2) choosing a different time-period; and (3) tripping transmission lines via a checklist

File details

The GitHub repository contains the following files required to run the application:

1. **Executable file (.exe)** - A 177MB standalone application that allows the users to run the Visayas Grid Model without installing Python or additional dependencies. It includes the graphical user interface and all built-in simulation functionalities.
2. **Documentation (PDF)** - Contains the User Manual, technical documentation, and test cases to provide guidance on installation, setup, and feature usage.
3. **Input Data Files (ZIP Archive)** - Contains CSV files for all network elements, such as: buses, generators, loads, transmission lines, shunts, transformers, and coordinates. This is used for modifying the network and creating new test scenarios, since the application is strict with formatting.

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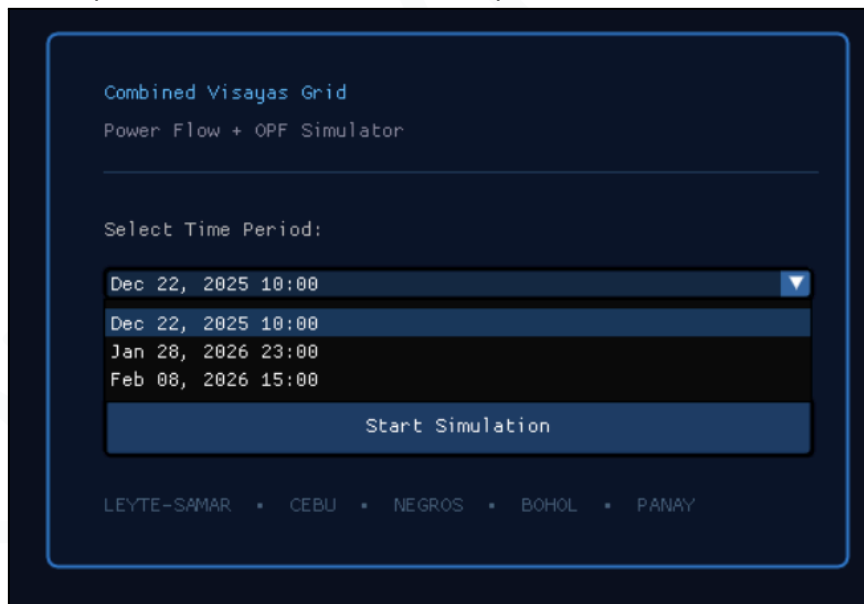
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Installation and Start-up

1. The application is a standalone executable. No Python installation or additional dependencies are required. Download the **Open Source Visayas Grid Model Simulator.exe** file and save it to a location of your choice. It is recommended to place the .exe file inside an empty folder, as running it will generate HTML files for the map and single-line diagram in the same directory.
2. Double-click **Open Source Visayas Grid Model Simulator.exe** to launch the application. A start-up window will appear.



3. Use the dropdown menu to select a time period.



4. Click **Start Simulation**. The application will run the power flow and OPF for the selected period.
5. Once complete, the results are displayed across the tabs in the main window. These will be described in detail in the succeeding section.

Result Tabs

After a simulation run, results are presented across four tabs in the main window. All tabs can be filtered by subgrid using the dropdown in the toolbar. The columns in each tab are described below.

Bus voltages and LMP

Displays the post-OPF state of every bus in the network with one row per bus.

Column	Description
Bus Name	Name of the bus as defined in the network data.
Subgrid	The island subgrid the bus belongs to.
Type	Bus type: SLACK (reference bus), PV (controlled voltage), or PQ (load bus).
Vn (kV)	Nominal voltage level of the bus in kilovolts.
Vm (pu)	Voltage magnitude from the power flow solution in per-unit.
Va (deg)	Voltage angle in degrees.
P (MW)	Net active power injection at the bus.
Q (MVar)	Net reactive power injection at the bus.
LMP (Php/MWh)	LMP at the bus from the OPF solution in Php/MWh.

Line loading

Displays power flow results for the transmission lines in the network, including tie-lines.

Column	Description
Line Name	Name of the bus as defined in the network data.
Type	The island subgrid the bus belongs to.
Subgrid	Bus type: SLACK (reference bus), PV (controlled voltage), or PQ (load bus).
From Bus	Nominal voltage level of the bus in kilovolts.
To Bus	Voltage magnitude from the power flow solution in per-unit.
Vn (kV)	Voltage angle in degrees.
Loading (%)	Net active power injection at the bus.
P_from (MW)	Net reactive power injection at the bus.
Q_from (MVar)	LMP at the bus from the OPF solution in Php/MWh.
I_from (kA)	Current at the sending end of the line.

Generator dispatch

Displays the optimised dispatch for every generating unit in the network, including the HVDC link chosen to be modelled as a generator.

Column	Description
Generator	Name of the generating unit as defined in the network data.
Plant Type	The plant technology.
Subgrid	The subgrid the generator is connected to.
Bus	Name of the bus the generator is connected to.
Dispatch (MW)	Optimal active power output determined by the OPF.
Q (MVar)	Reactive power output determined by the OPF.
Cost (Php/MWh)	The cost coefficient of the first segment of the generator's piecewise cost curve. For units with a single-segment, this is the unit's marginal cost.
LMP (Php/MWh)	LMP at the generator's connected bus.

Load LMP

Displays the active power consumption and LMP at each load bus across the network.

Column	Description
Load	Name of the load as defined in the network data.q
Subgrid	The subgrid the load is located in.
Bus	Name of the bus the load is connected to.
P (MW)	Active power consumed by the load from the OPF solution.
LMP (Php/MWh)	LMP at the load's bus representing the cost of energy at that location.

Modifying the Network

1. Network elements are defined through CSV files, with one file per component type per subgrid. To begin, duplicate the CSV file you wish to modify from the provided ZIP to ensure that the required format is preserved.
2. You may then edit the file by modifying existing rows to update elements, adding new rows for additional elements, or deleting rows to remove elements from the network.
3. Navigate to the **Modify Network** tab.
4. From the dropdown menus, select the appropriate subgrid and the component type that you intend to replace.
5. Before uploading, the application will display the currently stored CSV file in the default file preview, allowing you to review the existing data.

6. Upload your edited CSV file. Once uploaded the file will completely replace the existing component file for the selected subgrid.
7. After uploading, the updated CSV content will be reflected in the file preview. Note that these changes are applied within the application, while the original base data stored on disk remains unchanged. There is no required naming format for the uploaded CSV file.
8. Results before and after the network modification can be compared in the **Comparison Tab**, which is discussed in the succeeding sections.

Generators (*gens.csv*)

This defines all generating units connected to the subgrid. Each row represents one unit.

Column	Unit	Description
name	*	Name of the generating unit. Must match a corresponding entry in gen_costs.csv .
bus	*	The bus the generator is located in. It must match a bus name in buses.csv .
p_mw	MW	Must be a value greater than or equal to zero.

name	bus	p_mw
05CARMENDPP_U01	TOLEDO 2	0
05CARMENDPP_U02	TOLEDO 2	0
05CARMENDPP_U03	TOLEDO 2	0
05CARMENDPP_U04	TOLEDO 2	0
05CEDC_U01	DAANGLUNGSOD 2	61.57
05CEDC_U02	DAANGLUNGSOD 2	61.77
05CEDC_U03	DAANGLUNGSOD 2	63.1
05EAUC_U01	LAPU-LAPU 2	0
05EAUC_U02	LAPU-LAPU 2	7

⚠ NOTE ⚠

Adding or removing a generator requires a corresponding addition or removal of that unit's entry in **gen_costs.csv**. The two files must remain consistent as a generator present in one but not the other will cause the simulation to fail.

Loads (*loads.csv*)

Defines all load demands connected to the subgrid. Each row represents one load.

Column	Unit	Description
name	*	Name of the load.
bus	*	Name of the bus the load is connected to. Must match a bus name in buses.csv .
p_mw	MW	Must be a value greater than or equal to zero.

name	bus	p_mw
DAANB_SS	DAAN BANTAYAN	0
TALIS_T1L1	DAAN BANTAYAN 2	36.69
COMPS_SS	COMPOSTELA	0
COMPS_T1L1	COMPOSTELA 3	32.49
COMPS_T2L2	COMPOSTELA 3	61.28
CEBU_SS	CEBU	0
BANIL_T1L1	CEBU 3	66.82
BANIL_T1L2	CEBU 3	75.8
BANIL_T1L3	CEBU 3	75.66

Shunts (*shunts.csv*)

Defines capacitor banks and reactors connected to the network. Each row represents one shunt device.

Column	Unit	Description
name	*	Name of the shunt element.
bus	*	Name of the bus the shunt is connected to. Must match a bus name in buses.csv .
q_mvar	MVar	Enter a positive value for a capacitor bank (reactive power injection) and a negative value for a reactor or inductor (reactive power absorption).

name	bus	q_mvar
COMPST_C01	COMPOSTELA 2	-20
COMPST_C02	COMPOSTELA 2	-20
CEBU_C02	CEBU 2	-20
CEBU_C03	CEBU 2	-20
CEB69_C01	CEBU 3	-7.5
CEB69_C02	CEBU 3	-7.5
CEB69_C03	CEBU 3	-7.5
DAANB_S01	DAAN BANTAYAN	30
DAANB_S02	DAAN BANTAYAN	50

Generator costs (*gen_costs.csv*)

Defines the operating limits and piecewise-linear cost curve for each generating unit. Each row represents one unit.

Column	Unit	Description
name	*	Name of the generating unit. Must match the corresponding entry in gens.csv .
min_p_mw	MW	Minimum active power output.
max_p_mw	MW	This column copies the p_mw column of gens.csv . It does not affect the power flow simulation.
dep_capacity	MW	For most units this parameter reflects the declared effective

		maximum output that the unit can provide. For renewable energy units with variable output, this may be an exception.
min_q_mvar	MVar	Minimum reactive power output. This was calculated by using the formula: dep_cap * tan (arccos(-0.9)) . pf = 0.9 lead is from the set limits of a generating unit according to the Philippine Grid Code (GCR 4.4.2.1.3)
max_q_mvar	MVar	Maximum reactive power output. This was calculated by using the formula: dep_cap * tan (arccos(0.85)) . pf = 0.85 lag is from the set limits of a generating unit according to the Philippine Grid Code (GCR 4.4.2.1.3)
p1	MW	Start of the first cost segment. Typically 0 or the unit's minimum output.
p2	MW	End of the first cost segment. Typically equal to dep_capacity for single-segment units or a breakpoint for multi-segment units.
c1	Php/MWh	Cost coefficient for the first segment of the piecewise-linear cost curve. For single-segment units this is the unit's marginal cost. For multi-segment units, additional segment pairs (p3/p4/c2, p5/p6/c3) define progressively higher cost segments.

name	min_p_mw	max_p_mw	dep_capacity	min_q_mvar	max_q_mvar	p1	p2	c1	p3	p4	c2
05CARMENDPP_U01	0	0	10	-4.843221048	6.197443384	0	10	27587			
05CARMENDPP_U02	0	0	10	-4.843221048	6.197443384	0	10	27587			
05CARMENDPP_U03	0	0	10	-4.843221048	6.197443384	0	10	27587			
05CARMENDPP_U04	0	0	10	-4.843221048	6.197443384	0	10	27587			
05CEDC_U01	0	61.57	82	-39.7144126	50.81903575	0	45.1	-10000	45.1	83.2	3140
05CEDC_U02	0	61.77	82	-39.7144126	50.81903575	0	45.1	-10000	45.1	83.4	3140
05CEDC_U03	0	63.1	82	-39.7144126	50.81903575	0	45.1	-10000	45.1	83.1	3140
05CPPC_U01	0	0	6.5	-3.148093681	4.0283382	0	6.5	32000			
05CPPC_U02	0	0	6.5	-3.148093681	4.0283382	0	6.5	32000			
05CPPC_U03	0	0	6.5	-3.148093681	4.0283382	0	6.5	32000			
05CPPC_U04	0	0	6.5	-3.148093681	4.0283382	0	6.5	32000			
05CPPC_U05	0	0	6.5	-3.148093681	4.0283382	0	6.5	32000			
05CPPC_U06	0	0	6.5	-3.148093681	4.0283382	0	6.5	32000			
05CPPC_U07	0	0	6.5	-3.148093681	4.0283382	0	6.5	32000			

⚠ NOTE ⚠

The piecewise-linear cost curve is numerically sensitive. Segment breakpoints that are inconsistent or overlapping can cause the OPF to fail or produce unexpected LMP values. Always verify that segment endpoints are continuous.

Lines (*lines.csv*)

Defines all transmission lines within the subgrid. Each row represents one line. Parallel circuits between the same two buses are entered as separate rows.

Column	Unit	Description
name	*	Name of the line.
from_bus	*	Name of the sending-end bus. Must match a bus name in

buses.csv.		
to_bus	*	Name of the receiving-end bus. Must match a bus name in buses.csv .
length_km	km	Route length of the line in kilometers.
r_ohm_per_km	Ω/km	Resistance per unit length.
x_ohm_per_km	Ω/km	Reactance per unit length.
c_nf_per_km	nF/km	Line charging capacitance per unit length.
max_i_ka	kA	Thermal current rating of the line.

name	from_bus	to_bus	length_km	r_ohm_per_km	x_ohm_per_km	c_nf_per_km	max_i_ka
DAAN_COM1	DAAN BANTAYAN	COMPOSTELA	87.19	0.0794	0.248	10.6945385	1.2
DAAN_COM2	DAAN BANTAYAN	COMPOSTELA	87.19	0.0794	0.248	10.6945385	1.2
CEBU_COM1	COMPOSTELA 2	CEBU 2	15.32	0.0794	0.248	10.6945385	1.2
CEBU_COM2	COMPOSTELA 2	CEBU 2	15.32	0.0794	0.248	10.6945385	1.2
CEBU_MGD1	CEBU	MAGDUGO	33	0.0794	0.248	10.6945385	1.2
CEBU_MGD2	CEBU	MAGDUGO	33	0.0794	0.248	10.6945385	1.2
QUIO_CEB1	CEBU	QUIOT	10.5	0.0794	0.248	10.6945385	1.2
COLO_CEB2	CEBU	COLON	23.13	0.0794	0.248	10.6945385	1.2
COLO_CEB3	CEBU	COLON	23.13	0.0794	0.248	10.6945385	1.2
CEBU_MAN1	CEBU	MANDAUE	5.83	0.0794	0.248	10.6945385	1.2
CEBU_MAN2	CEBU	MANDAUE	5.83	0.0794	0.248	10.6945385	1.2
MAND_LAP1	MANDAUE	LAPU-LAPU	0.81	0.0794	0.248	10.6945385	1.2
MAND_LAP2	MANDAUE	LAPU-LAPU	0.81	0.0794	0.248	10.6945385	1.2
COLO QUI1	QUIOT	COLON	12.86	0.0794	0.248	10.6945385	1.2

Transformers (**transformers.csv**)

Defines all transformers in the subgrid. Each row represents one transformer unit.

Column	Unit	Description
name	*	Name of the transformer.
hv_bus	*	Name of the high-voltage side bus. Must match a bus name in buses.csv .
lv_bus	*	Name of the low-voltage side bus. Must match a bus name in buses.csv .
std_type	*	Name of the transformer standard type defining its electrical parameters. Must exactly match a name in std_types.csv . If the required type does not already exist, it must be added to the std_types.csv before uploading this file.

name	hv_bus	lv_bus	std_type
DAANB_TR1	DAAN BANTAYAN	DAAN BANTAYAN 2	150 MVA 230/69 kV
DAANB_TR2	DAAN BANTAYAN	DAAN BANTAYAN 2	100 MVA 230/69 kV
COMPST_TR1	COMPOSTELA	COMPOSTELA 2	150 MVA 230/138 kV
COMPST_TR2	COMPOSTELA	COMPOSTELA 2	150 MVA 230/138 kV
COMPST_TR3	COMPOSTELA 2	COMPOSTELA 3	100 MVA 138/69 kV
COMPST_TR4	COMPOSTELA 2	COMPOSTELA 3	100 MVA 138/69 kV
COMPST_TR5	COMPOSTELA	COMPOSTELA 2	150 MVA 230/138 kV
COMPST_TR6	COMPOSTELA	COMPOSTELA 2	150 MVA 230/138 kV
CEBU230_TR1	CEBU	CEBU 2	300 MVA 230/138 kV
CEBU230_TR2	CEBU	CEBU 2	300 MVA 230/138 kV
CEBU230_TR3	CEBU	CEBU 2	300 MVA 230/138 kV
CEBU_TR2	CEBU 2	CEBU 3	100 MVA 138/69 kV
CEBU_TR3	CEBU 2	CEBU 3	100 MVA 138/69 kV
CEBU_TR4	CEBU 2	CEBU 3	100 MVA 138/69 kV

Standard Transformer Types (*std_types.csv*)

Defines the electrical parameters for each transformer type referenced in **transformers.csv**. Each row represents one standard type. A type must be defined here before it can be used in **transformers.csv**.

Column	Unit	Description
name	*	Name of the standard type (e.g. 300 MVA 230/138 kV). This name is referenced in transformers.csv .
sn_mva	MVA	Rated apparent power of the transformer.
vn_hv_kv	kV	Rated voltage on the high-voltage side.
vn_lv_kv	kV	Rated voltage on the low-voltage side.
vk_percent	%	Short-circuit voltage as a percentage of rated voltage. Determines the transformer's series impedance.
vkr_percent	%	Representing resistive losses in the windings.
pfe_kw	kW	No-load or iron core losses at rated voltage.
i0_percent	%	No-load current as a percentage of rated current.
shift_degree	degrees	Phase shift introduced by the transformer winding configuration. Set to 0 for standard delta-wye or wye-wye transformers with no intentional phase shift.

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name	sn_mva	vn_hv_kv	vn_lv_kv	vk_percent	vkr_percent	pfe_kw	i0_percent	shift_degree
150 MVA 230/69 kV	150	230	69	16	0.533037284	50	0.1	0
100 MVA 230/69 kV	100	230	69	16	0.639488614	50	0.1	0
150 MVA 230/138 kV	150	230	138	16	0.533037284	50	0.1	0
300 MVA 230/138 kV	300	230	138	16	0.456956382	50	0.1	0
100 MVA 138/69 kV	100	138	69	12	0.47961646	50	0.1	0
50 MVA 138/69 kV	50	138	69	12	0.599251403	50	0.1	0
30 MVA 138/69 kV	30	138	69	12	0.798228126	50	0.1	0
37.5 MVA 138/34.5 kV	37.5	138	34.5	12	0.798228126	50	0.1	0
40 MVA 138/34.5 kV	40	138	34.5	12	0.599251403	50	0.1	0
25 MVA 69/13.8 kV	25	69	13.8	9	0.598671095	50	0.1	0
100 MVA 138/13.8 kV	100	138	13.8	12	0.47961646	50	0.1	0
150 MVA 138/13.8 kV	150	138	13.8	12	0.399777963	50	0.1	0
210 MVA 230/138 kV	210	230	18	16	0.456956382	50	0.1	0
38 MVA 138/13.8 kV	38	138	13.8	12	0.798228126	50	0.1	0

⚠ NOTE ⚠

If a transformer in **transformers.csv** references a standard type not present in this file, the simulation will fail. Always add new standard types here before uploading a modified **transformers.csv**.

Buses (*buses.csv*)

Defines all buses in the network. Each row represents one bus.

Column	Unit	Description
name	*	Name of the bus.
vn_kv	kV	Voltage level of the bus.

name	vn_kv
DAAN BANTAYAN	230
DAAN BANTAYAN 2	69
COMPOSTELA	230
COMPOSTELA 2	138
COMPOSTELA 3	69
CEBU	230
CEBU 2	138
CEBU 3	69
MANDAUE	138

⚠ NOTE ⚠

When adding a new bus, make sure to also add its corresponding geographical coordinates in **geo_coords.csv**. Likewise, when removing a bus, make sure to also remove its coordinates in the said CSV file.

Geographical Coordinates (*geo_coords.csv*)

Maps each bus to a geographical location used to plot substations on the interactive map. Every bus in **buses.csv** should have a corresponding entry here. Buses without coordinates will not appear on the map.

Column	Unit	Description
bus	*	Name of the bus. Must match a bus name in buses.csv .
lat	Decimal degrees	Latitude of the substation.
lng	Decimal degrees	Longitude of the substation.

bus	lat	lng
DAAN BANTAYAN	11.252494	124.066132
DAAN BANTAYAN 2	11.252494	124.066132
COMPOSTELA	10.466319	124.006533
COMPOSTELA 2	10.466319	124.006533
COMPOSTELA 3	10.466319	124.006533
CEBU	10.3583891	123.9190058
CEBU 2	10.3583891	123.9190058
CEBU 3	10.3583891	123.9190058
MANDAUE	10.329651	123.963674

⚠ NOTE ⚠

Multiple buses at the same substation (e.g. different voltage levels) should share the same coordinates. They will be rendered as a single marker on the map with tabbed voltage levels in the popup.

Simulating Line Outages

The **Line Outages** tab allows you to simulate contingency scenarios by taking one or more transmission lines out of service.

Setting the outages

1. Navigate to the **Line Outages** tab. Note that the line list will only appear after at least one simulation has been run.
2. Use the **Search** box to filter lines by name, or use the Subgrid dropdown to filter by island. Both filters can be used simultaneously.
3. Tick the checkbox next to the line/s you wish to trip. Checked lines are highlighted in red. Multiple lines may be selected at once.

2 line(s) tripped: KANA_ORM1 976 MVA, KANA_ORM2 976 MVA

☐ [LEYTE-SAMAR] KANA_TAB1 488 MVA
☐ [LEYTE-SAMAR] KANA_TAB2 488 MVA
☒ [LEYTE-SAMAR] KANA_ORM1 976 MVA
☒ [LEYTE-SAMAR] KANA_ORM2 976 MVA
☐ [LEYTE-SAMAR] CBANK_CTP1 63MVA
☐ [LEYTE-SAMAR] CTAP_SNA1 63MVA
☐ [LEYTE-SAMAR] CLBY_CTP1 63 MVA
☐ [LEYTE-SAMAR] CLBY_PAR1 196MVA

4. Click **Run Simulation** from the top toolbar to apply the outages. Lines are not taken out of service until the simulation is run.
5. Once the simulation completes, all result tabs will reflect the network with the selected line/s removed. The status bar will confirm the number of active outages. Tripped lines are also visually marked on the map, which is discussed further in the next section.
6. Results before and after the tripped lines can be compared in the **Comparison Tab**, which is discussed in the succeeding sections.

Clearing outages

1. To remove a specific outage, uncheck its checkbox and re-run the simulation.
2. To clear all active outages at once, click **Clear All Outages** and re-run the simulation.
3. If the line list appears outdated, click **Refresh List** to reload it from the most recent simulation.

⚠ NOTE ⚠

Certain combinations of simultaneously tripped lines may cause the OPF to fail, as the solver may not find a solution within the set network limits. If this occurs, consider adjusting the simulation constraints in the **Settings** tab, which is discussed in the next section.

```

Simulation Error
Traceback (most recent call last):
  File "Visayas Map.py", line 1286, in run_simulation
    _LOAD_STEM = 12 # stem: bar/rail ? triangle top
    ~~~~~
  File "pandapower\run.py", line 478, in runopp
    ~~~~~
  File "pandapower\optimal_powerflow.py", line 76, in _optimal_powerflow
    raise OPFNotConverged("Optimal Power Flow did not converge!")
pandapower.auxiliary.OPFNotConverged: Optimal Power Flow did not converge!

??? HINT ?????????????????????????????????????????????
OPF did not converge. Common causes:
* Cost curve segments are infeasible
* p_mw outside [min_p_mw, max_p_mw]
* Total generation cannot meet total load

Click Reset Overrides to restore the default network.

Close
    
```

Comparison Tab

The **Comparison Tab** allows you to compare results between any two simulation runs. This helps the user evaluate the effect of different time periods, network

modifications, or line outage scenarios side by side. Differences are color-coded, where green indicates an improvement relative to the baseline and red indicates a worsening.

Setting the Baseline

1. Run the first scenario you wish to use as your reference. This can be any period, network modification, outage configuration, or both.
2. Once the simulation completes, click **Pin as Baseline** in the top toolbar. This saves the current results as the baseline.
3. Change the period, apply network modifications, set line outages, or both to define your second scenario.
4. Run the simulation again. The results of this run are automatically stored as the modified scenario.
5. Before and after results can be compared in the **Comparison Tab**, which is discussed in the next section.

Viewing the Comparison

1. Navigate to the **Comparison Tab**. The header will display the labels of the baseline and modified runs for reference.
2. Use the inner tabs to switch between **Bus Voltages & LMP**, **Line Loading**, **Generator Dispatch**, and **Load LMP** comparisons. The parameters shown in each tab are the same as those discussed in the previous sections.
3. Use the **Subgrid** dropdown in the top toolbar to filter results by island.

Compare any two runs across periods, overrides, or outages.
 1. Run any period/scenario 2. Click 'Pin as Baseline' 3. Run another 4. View diff here.
 Green = improved vs baseline, Red = worsened.

Baseline: Dec 22, 2025 10:00
 Modified: Dec 22, 2025 10:00 + 1 outage(s)

Bus Voltages & LMP Line Loading **Generator Dispatch** Load LMP

Generator	Subgrid	Bus	P base(MW)	P mod(MW)	Q base(Mvar)	Q mod(Mvar)	LMP base(PHP)	LMP mod(PHP)
04CLBYBNK_G01	LEYTE-SAMAR	CALBAYOG BUNKER 2	0.000	0.000	-1.163	-0.899	2879.03	2882.01
04IASMOD_G01	LEYTE-SAMAR	ISABEL 1	0.000	0.000	-4.009	-3.202	2773.55	2776.42
04IASMOD_G02	LEYTE-SAMAR	ISABEL 1	0.000	0.000	-4.134	-3.245	2773.55	2776.42
04IASMOD_G03	LEYTE-SAMAR	ISABEL 1	0.000	0.000	-6.452	-5.421	2773.55	2776.42
04IASMOD_G04	LEYTE-SAMAR	ISABEL 1	0.000	0.000	-4.100	-3.287	2773.55	2776.42
04IASMOD_G05	LEYTE-SAMAR	ISABEL 1	0.000	0.000	-6.452	-5.421	2773.55	2776.42
04IASMOD_G06	LEYTE-SAMAR	ISABEL 1	0.000	0.000	-4.100	-3.287	2773.55	2776.42
04ISIDROSOL_G01	LEYTE-SAMAR	TABANGO 2	51.450	51.450	-48.674	-48.674	2802.33	2833.44
04LEYTE_A	LEYTE-SAMAR	KANANGA	306.000	306.000	-91.564	-81.277	2774.08	2773.12
04LGPP_G01	LEYTE-SAMAR	TONGONAN 3	79.500	79.500	-27.540	-21.384	2746.44	2749.36

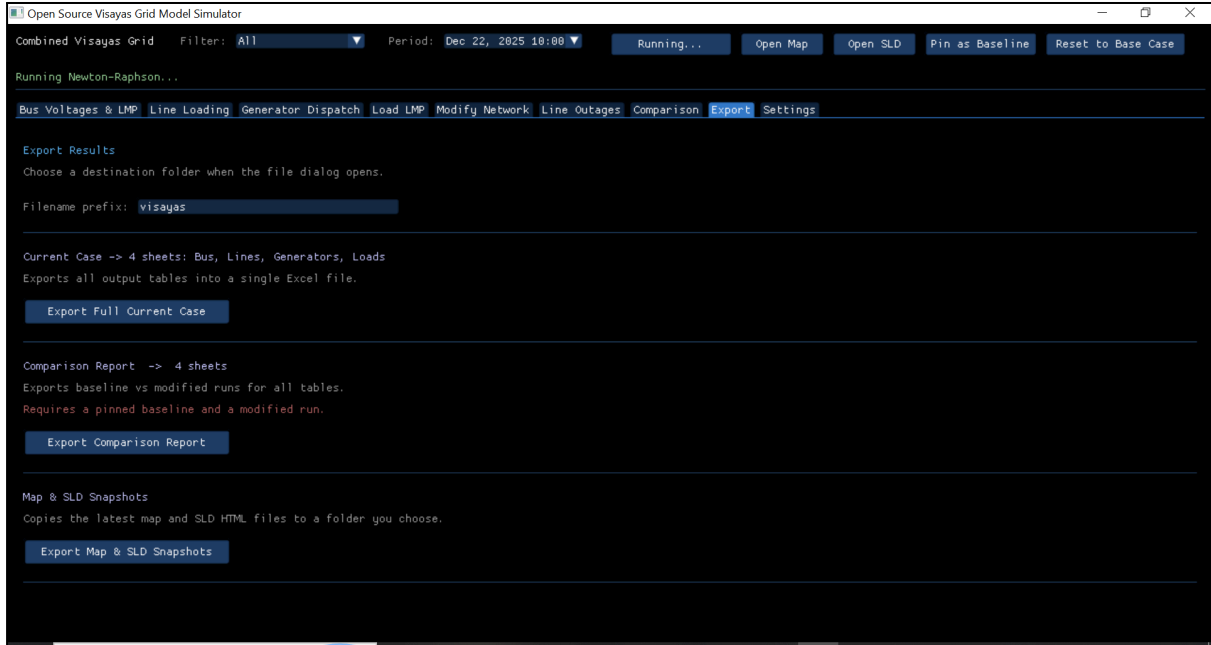
Export Tab

The **Export Tab** allows you to save simulation results to your local directory. Before exporting, enter a desired prefix in the **Filename prefix** field.

1. To export all output tables as a single Excel file, click **Export Full Current Case**.
2. To export a side-by-side comparison of a baseline and modified run, first pin a baseline by clicking **Pin as Baseline** from the top toolbar after running the simulation, then run a subsequent simulation with your modifications applied. Click **Export Comparison Report** to export this.

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3. To save the latest map and SLD as HTML files, click **Export Map & SLD Snapshots**.
4. For each export, a file dialog will open where you can choose the destination folder and specify the filename for the exported file.



Modifying Simulation Constraints

The **Settings Tab** allows you to adjust the solver and network parameters used. There are cases where the OPF may fail due to the set simulation constraints, such as when simulating network modifications or line outages, in which case these settings can be adjusted accordingly to help the solver converge.

1. Under **Voltage Limits**, set the minimum and maximum allowable bus voltages in the Min Vm (pu) and Max Vm (pu) fields. The defaults are 0.95 pu and 1.05 pu, respectively.
2. Under **Power Flow (Newton Raphson)**, set the maximum number of iterations allowed for the OPF solver in its Max Iterations field. The default is 100.
3. Under **Optimal Power Flow**, set the maximum number of iterations allowed for the OPF solver in its Max Iterations field. The default is 100.
4. Under **Tie Line Parameters**, set the resistance, reactance, and capacitance per kilometer of the tie line in the R (Ω/km), X (Ω/km), and C (nF/km) fields. The defaults are 0.0277, 0.1200, and 110.0 respectively.
5. Use the - and + buttons to adjust any value, or type directly into the fields.
6. Clicking **Reset All to Defaults** to restore all fields to their original values.

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Visayas Map

The map provides a geographic view of the Visayas grid. To open it, click **Open Map** in the toolbar after a simulation has been run.

1. Hover over any element to view basic information. Click to open the full details panel.
2. Substations with multiple voltage levels are displayed as a single marker. The size of the marker is scaled according to the generator or load at that substation. Click the substation and use the voltage level tabs to switch between levels.
3. Clicking a substation opens a details panel showing the LMP at that bus, as well as a Generator Dispatch summary listing each unit, its type, and its dispatched output in MW. A Load Demand summary is also shown listing each load and its demand in MW.
4. Hover over a line to view its name and loading percentage. For those with parallel lines, the details panel will show a tab for each circuit.
5. The lines connecting islands are shown as dashed lines. HVDC links are shown as thick dashed lines with arrows on both endpoints. Click an HVDC link to view its full details.
6. Outaged lines are shown with a black dashed appearance and a pulsing animation to distinguish them from in-service lines.
7. A legend at the lower left corner indicates bus and line types. The layer panel at the upper right contains checkboxes to show or hide individual layers (subgrids, tie lines, HVDC interconnections, outage lines, generator dispatch, and load demand).
8. Below the layer panel is the line color mode selector.

Visayas Single-Line Diagram

The Single-Line Diagram (SLD) provides a schematic view of the Visayas network. To open it, click **Open SLD** in the top toolbar after a simulation has been run. This will open an HTML page in a browser containing the full diagram.

1. Hover over any element to view its name and key parameters.
2. For parallel lines sharing the same corridor, each circuit is drawn as a separate offset path.

3. Each subgrid is enclosed in a labeled dashed boundary box for easy identification.
4. Scroll to zoom in and out. Click and drag to pan across the diagram. Click **Reset** in the top right corner to return to the default view.
5. A top ribbon displays a quick reference for voltage levels and element symbols.
6. A detailed legend on the right side of the diagram indicates voltage levels, line loading, bus types, and element symbols.

⚠ **NOTE** ⚠

Any modifications to the network such as adding or removing generators, loads, or other components will be reflected in the SLD after clicking **Run Simulation**. However, significant changes to the network may affect the diagram layout and cause elements to shift or appear misaligned.

Support

For questions, issues, or feedback regarding the application, please reach out to the team through the following:

Name	Email
Denzel Andre Alvaro	dbalvaro@up.edu.ph
Mia Therese Domaboc	mtdomaboc@up.edu.ph
Roland Jacob Orengo	rcorengo1@up.edu.ph